Statistics & Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Data representation**

The bar chart shows the distribution of grades in a class of 37 students. Find the probability that a randomly selected student earned a grade of A or B, as a percent to the nearest whole percent.

$P(A \text{ or } B) = \frac{8+15}{37} = \frac{23}{37} \approx 0.622$, or about 62%.

2 **[Foundational]** • **Target: Data representation**

Using the same bar chart, find the probability that a randomly selected student scored a C or D grade, as a percent to the nearest whole percent.

$P(C \text{ or } D) = \frac{10+4}{37} = \frac{14}{37} \approx 0.378$, or about 38%.

3 **[Foundational]** • **Target: Permutations**

How many ways can 5 distinct books be arranged on a shelf?

$5! = 120$.

4 **[Foundational]** • **Target: Permutations**

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?

${}_{12}P_3 = 12 \times 11 \times 10 = 1320$.

5 **[Foundational]** • **Target: Permutations**

How many distinct arrangements are there of the letters in the word APPLE?

There are 5 letters with the letter P repeated twice: $\frac{5!}{2!} = \frac{120}{2} = 60$.

6 **[Foundational]** • **Target: Combinations**

How many ways can a committee of 4 be chosen from 11 people?

$\binom{11}{4} = \frac{11!}{4!\,7!} = 330$.

7 **[Foundational]** • **Target: Combinations**

A pizza shop offers 8 toppings. How many different 3-topping pizzas can be made?

$\binom{8}{3} = \frac{8 \times 7 \times 6}{6} = 56$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Combinations**

From a group of 6 men and 5 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?

$\binom{6}{2}\binom{5}{2} = 15 \times 10 = 150.$

9 **[Intermediate]** • **Target: Probability rules**

A fair die is rolled. Find $P(\text{even or greater than } 4)$.

Even $= \{2, 4, 6\}$, greater than $4 = \{5, 6\}$. Union $= \{2, 4, 5, 6\}$. $P = \frac{4}{6} = \frac{2}{3}$.

10 **[Intermediate]** • **Target: Probability rules**

$P(A) = 0.45$ and $P(B) = 0.3$, and A and B are mutually exclusive. Find $P(A \text{ or } B)$.

$P(A \text{ or } B) = 0.45 + 0.3 = 0.75.$

11 **[Intermediate]** • **Target: Probability rules**

A card is drawn from a standard 52-card deck. Find $P(\text{not a face card})$.

There are 12 face cards, so 40 are not face cards. $P = \frac{40}{52} = \frac{10}{13}$.

12 **[Intermediate]** • **Target: Conditional probability and independence**

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?

$P(A)P(B) = 0.5 \times 0.4 = 0.2 = P(A \cap B)$. Yes, A and B are independent.

13 **[Intermediate]** • **Target: Conditional probability and independence**

In a class, 60% of students play sports, and 25% of all students both play sports and are honor students. Find $P(\text{honor student} \mid \text{plays sports})$.

$P(\text{honor} \mid \text{sports}) = \frac{0.25}{0.6} \approx 0.417$, or about 41.7%.

14 **[Intermediate]** • **Target: Conditional probability and independence**

Events C and D are independent, with $P(C) = 0.7$ and $P(D) = 0.6$. Find $P(C \cup D)$.

$P(C \cap D) = 0.7 \times 0.6 = 0.42$. $P(C \cup D) = 0.7 + 0.6 - 0.42 = 0.88.$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Measures of center and spread • Cross-Lesson: 1.01 Advanced Algebra

Find the mean and median of the data set $12, 15, 18, 20, 25$.

Mean $= \frac{12+15+18+20+25}{5} = \frac{90}{5} = 18$. Median (middle value, already sorted) $= 18$.

16 [Challenging] • Target: Measures of center and spread • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Find the population standard deviation of the data set $4, 8, 6, 5, 3$.

Mean $= \frac{26}{5} = 5.2$. Squared deviations: $1.44, 7.84, 0.64, 0.04, 4.84$, summing to 14.8. Variance $= \frac{14.8}{5} = 2.96$. Standard deviation $= \sqrt{2.96} \approx 1.72$.

17 [Challenging] • Target: The binomial theorem • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the coefficient of x^3y^2 in the expansion of $(x+y)^5$.

The term is $\binom{5}{2}x^3y^2$, so the coefficient is $\binom{5}{2} = 10$.

18 [Challenging] • Target: The binomial theorem • Cross-Lesson: 4.01 Lines, Triangles & Polygons

In the expansion of $(2x-1)^4$, find the coefficient of x^2.

The general term is $\binom{4}{k}(2x)^{4-k}(-1)^k$. For x^2, $4-k=2 \Rightarrow k=2$: $\binom{4}{2}(2x)^2(-1)^2 = 6(4x^2) = 24x^2$. The coefficient is 24.

19 [Challenging] • Target: The normal distribution • Cross-Lesson: 5.01 Trigonometry

A data set is normally distributed with mean 100 and standard deviation 15. What percent of values lie between 70 and 130?

70 and 130 are each 2 standard deviations from the mean (100 ± 30). By the 68-95-99.7 rule, about 95% of values lie in this range.

20 [Challenging] • Target: The normal distribution • Cross-Lesson: 6.01 Circles & Coordinate Geometry

For the same distribution (mean 100, standard deviation 15), find the approximate percentage of values greater than 115.

115 is 1 standard deviation above the mean. About 68% of values lie within 1 standard deviation, so 34% lie between the mean and 115. Since 50% lie above the mean, the percentage above 115 is $50\% - 34\% = 16\%$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Permutations

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 8 members?

a. 512

b. 330

c. 342

d. 336 ✓

${}_8P_3 = 8 \times 7 \times 6 = 336$.

22 [Foundational] • Target: Combinations

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 16 members?

a. 3360 ✓

b. 4096

c. 3366

d. 3354

${}_{16}P_3 = 16 \times 15 \times 14 = 3360$.

23 [Foundational] • Target: Probability rules

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 10 members?

a. 714

b. 726

c. 720 ✓

d. 1000

${}_{10}P_3 = 10 \times 9 \times 8 = 720$.

24 [Foundational] • Target: Conditional probability and independence

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 20 members?

a. 6834

b. 6840 ✓

c. 6846

d. 8000

${}_{20}P_3 = 20 \times 19 \times 18 = 6840$.

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25 [Foundational] • Target: Measures of center and spread

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 9 members?

a. 504 ✓

b. 498

c. 729

d. 510

${}_9P_3 = 9 \times 8 \times 7 = 504$.

26 [Foundational] • Target: Data representation

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 6 members?

a. 216

b. 114

c. 120 ✓

d. 126

${}_6P_3 = 6 \times 5 \times 4 = 120$.

27 [Foundational] • Target: The binomial theorem

In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 11 members?

a. 996

b. 984

c. 1331

d. 990 ✓

${}_{11}P_3 = 11 \times 10 \times 9 = 990$.

28 [Foundational] • Target: The normal distribution

How many ways can a committee of 5 be chosen from 11 people?

a. 465

b. 459

c. 462 ✓

d. 467

$\binom{11}{5} = 462$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: quality control sampling

How many ways can a committee of 5 be chosen from 19 people?

a. 11631

b. 27132

c. 11628 ✓

d. 11625

$\binom{19}{5} = 11628.$

30 [Intermediate] • Target: Counting Principles & Probability

How many ways can a committee of 4 be chosen from 17 people?

a. 2383

b. 6188

c. 2380 ✓

d. 2377

$\binom{17}{4} = 2380.$

31 [Intermediate] • Target: Conditional Probability & Independence

How many ways can a committee of 4 be chosen from 20 people?

a. 4842

b. 4845 ✓

c. 4848

d. 15504

$\binom{20}{4} = 4845.$

32 [Intermediate] • Target: Normal Distributions & Standard Scores

How many ways can a committee of 5 be chosen from 16 people?

a. 8008

b. 4371

c. 4365

d. 4368 ✓

$\binom{16}{5} = 4368.$

33 [Intermediate] • Target: Regression & Correlation

How many ways can a committee of 4 be chosen from 7 people?

a. 21

b. 32

c. 38

d. 35 ✓

$\binom{7}{4} = 35.$

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34 [Intermediate] • Target: Permutations

Events C and D are independent, with $P(C) = 0.2$ and $P(D) = 0.9$. Find $P(C \cup D)$.

a. 0.97

b. 0.82

c. 0.92 ✓

d. 1.02

$P(C \cap D) = 0.2 \times 0.9 = 0.18$. $P(C \cup D) = 0.2 + 0.9 - 0.18 = 0.92$.

35 [Intermediate] • Target: Combinations

Events C and D are independent, with $P(C) = 0.3$ and $P(D) = 0.3$. Find $P(C \cup D)$.

a. 0.56

b. 0.41

c. 0.61

d. 0.51 ✓

$P(C \cap D) = 0.3 \times 0.3 = 0.09$. $P(C \cup D) = 0.3 + 0.3 - 0.09 = 0.51$.

36 [Intermediate] • Target: Probability rules

Events C and D are independent, with $P(C) = 0.6$ and $P(D) = 0.6$. Find $P(C \cup D)$.

a. 0.94

b. 0.89

c. 0.84 ✓

d. 0.74

$P(C \cap D) = 0.6 \times 0.6 = 0.36$. $P(C \cup D) = 0.6 + 0.6 - 0.36 = 0.84$.

37 [Intermediate] • Target: Conditional probability and independence

Find the coefficient of x^7y^3 in the expansion of $(x + y)^{10}$.

a. 118

b. 122

c. 120 ✓

d. 210

The term is $\binom{10}{7}x^7y^3$, so the coefficient is $\binom{10}{7} = 120$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Measures of center and spread • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Find the coefficient of x^6y^3 in the expansion of $(x+y)^9$.

a. 82

b. 126

c. 86

d. 84 ✓

The term is $\binom{9}{6}x^6y^3$, so the coefficient is $\binom{9}{6} = 84$.

39 [Challenging] • Target: Data representation • Cross-Lesson: 5.01 Trigonometry

From a group of 10 men and 8 women, how many ways can a committee of 5 be formed with exactly 3 men and 2 women?

a. 148

b. 3350

c. 3370

d. 3360 ✓

$\binom{10}{3}\binom{8}{2} = 120 \times 28 = 3360$.

40 [Challenging] • Target: The binomial theorem • Cross-Lesson: 6.01 Circles & Coordinate Geometry

From a group of 7 men and 7 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?

a. 42

b. 441 ✓

c. 431

d. 451

$\binom{7}{2}\binom{7}{2} = 21 \times 21 = 441$.

41 [Challenging] • Target: Counting Principles & Probability • Cross-Lesson: 7.01 Solid Geometry & Similarity

A password consists of 3 letters followed by 2 digits. If repetition of letters is NOT allowed, how many such passwords can be created?

a. $1,560,000$ ✓

b. $1,757,600$

c. $1,404,000$

d. $2,600,000$

Letter arrangements: $26 \times 25 \times 24 = 15,600$. Digit arrangements: $10 \times 10 = 100$. Total $= 15,600 \times 100 = 1,560,000$.

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42 [Challenging] • Target: Conditional Probability & Independence • Cross-Lesson: 1.01 Advanced Algebra

If $P(A) = 0.4$, $P(B \mid A) = 0.75$, and $P(B) = 0.5$, what is $P(A \mid B)$?

a. 0.60 ✓

b. 0.50

c. 0.30

d. 0.75

$P(A \cap B) = P(A)P(B \mid A) = 0.4 \times 0.75 = 0.3$. Then $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.60$.

43 [Challenging] • Target: Conditional Probability & Independence • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In a certain school, 60% of students take French, 40% take Spanish, and 20% take both. What is the probability that a student takes French given that they take Spanish?

a. 0.50 ✓

b. 0.33

c. 0.60

d. 0.25

$P(\text{French} \mid \text{Spanish}) = \frac{P(\text{French} \cap \text{Spanish})}{P(\text{Spanish})} = \frac{0.20}{0.40} = 0.50$.

44 [Challenging] • Target: Normal Distributions & Standard Scores • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A dataset has mean $\mu = 100$ and variance $\sigma^2 = 25$. What score corresponds to a z-score of 1.8?

a. 109 ✓

b. 105

c. 118

d. 145

Standard deviation $\sigma = \sqrt{25} = 5$. Score $x = \mu + z\sigma = 100 + 1.8(5) = 100 + 9 = 109$.

45 [Challenging] • Target: Regression & Correlation • Cross-Lesson: 4.01 Lines, Triangles & Polygons

If the correlation coefficient between two variables is $r = -0.85$, what does this indicate?

a. **A strong negative linear association** ✓

b. A weak negative linear association

c. A strong positive linear association

d. No linear association

Values of r near -1 indicate a strong negative linear relationship between the two variables.

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